

User manual

In order to use the ticket managing system, the user only has to remember their UID. The user does NOT have to remember other information, for example, an organizer does NOT have to remember the EID of the events they created.

Menu

```
1 Operations
2 Queries
3 Reports
4 Organizer toolkit (help you decide price tiers)
5 Exit
Choose by entering a number.
```

The user chooses what to do by typing the code of that operation. For example, if the user wants to go to “Reports” sub-menu, they should type “3”.

```
1.1 Rank number of tickets sold & gross revenue by city.
1.2                               by venue within a city.
2.1 Report number of events & performances by segment and genre.
2.2                               by country.
2.3                               by country and city.
2.4                               by country, city and venue.
3.1 Rank organizers by overall revenue.
3.2                               by country revenue.
3.3                               by country, city revenue.
4 Report ticket scalpers.
5.1 Rank customers by number of orders.
5.2                               by number of orders per city.
6.1 Rank customers by number of tickets cancelled.
6.2 Rank organizers by number of performances cancelled.
7 Report performance sell-through rates.
8.1 Report event number of completed resales.
8.2                               average markup over face value.
8.3                               fraction of listings priced exactly at the cap.
8.4 Report top 10 events by resale volume in a given period.
9 Report event comment most popular noun phrase.
Choose by entering a choice, e.g. 1.1
```

Similarly, if the user wants to “Rank number of tickets sold & gross revenue by city”, they should type “1.1”.

```
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Choose by entering a number.
1
1 Create a profile.
2 Delete a profile.
3 Create an event.
4 Create a performance.
5 Define pricetiers and assign to sections.
6 Set resale cap of an event.
7 Update price tier price.
8 Block or unblock seat.
9 Book tickets.
10 Cancel performance.
11 Cancel ticket.
12 List ticket.
13 Withdraw listing.
14 Purchase listing of another customer.
15 Review performance.
Choose by entering a number.
1
Enter name:
Yuhan Fan
Enter address:
sample address
Enter email:
sample@mail.utoronto.ca
Enter date-of-birth: YYYY-MM-DD
2005-03-04
```

```

Are you a customer or organizer? c/o
c
Enter payment card number:
1234123412345678
Your profile is created. UID: 106
1 Operations
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```

Create a profile

To begin with, create your profile. It is under “1 Operations” and “1 Create a profile”. You have to decide whether you are a customer or an organizer. The system will give you a UID, and that is what you have to remember in order to use the system.

Other operations

For “Operations”, the system gathers UID first, and will output useful information first and ask for further user response. For example, if an organizer wants to update a price tier, the system will: 1) ask for their UID, 2) output all the events they have created, 3) ask for EID, 4) output all the performances of that event, 5) ask for PID, 6) output all the price tiers of that performance, 7) ask for price tier name and new price. That is why the user does not have to remember things other than their UID.

```

Enter your UID:
1
Here are the events you have created:
EID      Event name
-----
1        Northern Lights Rock Night
2        Midnight Jazz Sessions
3        Electric Horizon Festival
4        Golden Country Evening
Enter EID:
1
Here are the performances of this event:
PID      Datetime      Venue
-----
1        2026-03-06 19:30:00  HISTORY
2        2026-07-11 20:00:00  HISTORY
3        2026-10-18 19:30:00  HISTORY
Enter PID:
3
Here are the price tiers of this performance:
Price tier name      Price
-----
Standard            40.00
VIP                 85.00
Which price tier do you want to update? Enter the name:
Standard
Enter the new price:
42
Sorry, since there are tickets sold in this price tier, you cannot update the price.

```

However, sometimes the system may directly ask the customer for PID after gathering UID. That is because the customer should have found the PID and performance information from the system before they do operations like placing an order or purchasing a listing.

Organizer toolkit

This is a useful service for new organizers. After creating a performance, they can ask the system to suggest a price tiers structure, and assignment of price tiers to sections. Using the organizer toolkit does not create the price tiers, nor does it assign to sections. The organizer has to go to “1 Operations” and then “5 Define pricetiers and assign to sections”.

Important rules

When your operation fails, you can either read the error message, or you may check here to see which rule it violates.

For all users: One has to be older than 18 to create a profile. When you respond to the system, follow the format it requires.

For customers: Tickets can only be booked for future non-cancelled performances. Tickets can only be cancelled more than 7 days before the performance. A reserved ticket cannot be booked if the seat is sold or blocked. A general ticket cannot be booked if the section is full. A listing cannot exceed the resale price cap of the event. A customer may only review a performance once, in a month after the performance, if they hold a ticket or listing that is not sold or for resale.

For organizers: Only reserved seats can be blocked. Seats that are sold cannot be blocked. The price of a price tier can only be changed if there is no uncancelled ticket placed on any sections of that price tier.

System limitations and future improvements

This system currently does not support concurrent access. To turn it into a useful ticket managing system in real life, synchronization must be added to support concurrent access.

This system interacts with the user via the terminal, and does not have any visualization. To put into use in real life, an application or a website should be built on top of it, allowing visualization like seat maps.